Islamic University, Kushtia
Department of Statistics and Data Science
B. Sc (Hons.) 1st Year 2nd semester Practical Examination-2025
Course no: CSE-1207, Course Title: Programming with C/C++

Answer 3(three) questions including question no. 5

Full Marks: 70	Time: 6 Hours	Type: Lab

1. Topic: Central Tendency

Dataset

Download the dataset from the link below:

https://drive.google.com/file/d/1NdtJoD-7D-NsfsE8gKJh-KujgM63Zb_f/view?usp=sharing

The dataset contains 9,994 rows and the following 21 columns. Relevant columns are:

Column	Type	Description
Sales	float	Revenue for the line item, in USD (target column for this task)
Quantity	int	Units sold in the line item
Discount	float	Discount applied, as a fraction (0 to 1)
Profit	float	Profit for the line item, in USD (can be negative)

Question

Write a complete C program that:

1. Reads sales_dataset.csv
2. Computes the *Arithmetic Mean*, *Geometric Mean*, and *Harmonic Mean* of all valid Sales values.
3. Sorts the valid Sales values and computes the *Median* and *Mode* (report the smallest value among any ties for mode).
4. Prints the number of valid rows used, the number of rows skipped, and all five statistics, each to 2 decimal places.

2. Topic: Mesures of Dispersion

Dataset

Download the dataset from the link below:

https://drive.google.com/file/d/1gTDloRXTbTYbv9uZFMlaFNsLQRdoge-V/view?usp=sharing

The dataset contains 49,222 rows and the following 17 columns:

Column	Type	Description
CustomerID	int	Unique identifier for the customer who placed the order
Quantity	int	Number of units purchased in the order
Age	int	Age of the customer
UnitPrice	int	Price per unit of the product — target column for this analysis
Sales	int	Sales amount for the order line

For this question, the column of interest (the target variable for all dispersion calculations) is UnitPrice.

Question

A business analyst wants to describe how spread out the UnitPrice values are across all orders in the dataset above. Using the dataset provided, write a complete C program that:

1) Reads the dataset from the CSV file and extracts the UnitPrice column into an array.
2) Sorts the data (needed for the quartile-based measures).
3) Computes and prints the following measures of dispersion:
   a. Range
   b. Variance
   c. Standard Deviation (SD)
   d. Mean Deviation (MD) about the mean
   e. Quartile Deviation (QD)
   f. Interquartile Range (IQR)
   g. Coefficient of Range
   h. Coefficient of Variation (CV)
   i. Coefficient of Mean Deviation about the mean
   j. Coefficient of Quartile Deviation

Prints every result clearly labelled, with an appropriate number of decimal places

3. Topic: Moments

Dataset

Download the dataset from the link below:

https://drive.google.com/file/d/1cy9-g1r4O7c3H1Tm_4RiLd6vuhlqtZ6N/view?usp=sharing

The dataset contains 1,000 rows and the following 4 columns:

Column	Type	Description
Gender	string	Student's gender (Male / Female)

Attendance	float	Class attendance percentage (2nd column in the CSV — default analysis column for this assignment)
Hours_Studied	float	Hours the student spent studying
Exam_Score	float	Final exam score

Question

A researcher wants to describe the shape of the distribution of a chosen numeric column from the dataset above (by default, Attendance) using raw moments and central moments. Using the dataset provided above, answer the following:

Write a complete C program that:

1. Read the dataset from a CSV file.
2. Extract a single numeric column from the CSV.
3. Compute the first four raw moments (μ1′, μ2′, μ3′, μ4′) and central moments μ2, μ3, and μ4 of the extracted column,
4. Compute and print the coefficient of skewness and the coefficient of kurtosis along with the related quantities $\beta 1 = \gamma 1^2$ and excess kurtosis $\gamma 2 = \beta 2 - 3$.
5. Print interpretation of the skewness (approximately symmetric, moderately/highly right- or left-skewed) and kurtosis (mesokurtic, leptokurtic, or platykurtic) based on the computed values.

4. Topic: Correlation

Dataset

Download the dataset from the link below:
https://drive.google.com/file/d/1XeFUtNFsnoHdscR1OgB5fMMNimvJx2pk/view?usp=drivesdk
The dataset contains the following 2 columns:

Column	Type	Description
Temperature	float	Temperature recorded during the sales period, in degrees Celsius
Revenue	float	Ice cream sales revenue corresponding to each temperature, in USD

Question

A researcher wants to measure the strength and direction of the linear relationship between Temperature and Revenue. Using the dataset provided above, answer the following:

Write a complete C program that:

1. Reads the dataset from the CSV file, using the header row to locate the requested columns by name.
2. Parses each row, skipping (and counting) any row where a value is missing or non-numeric.
3. Computes the Pearson correlation coefficient (r) between Temperature and Revenue using the standard formula:
   $r = [\, n\Sigma xy - (\Sigma x)(\Sigma y) \,] \,/\, sqrt\{\, [\, n\Sigma x^2 - (\Sigma x)^2 \,][\, n\Sigma y^2 - (\Sigma y)^2 \,] \,\}$
4. Prints the number of valid rows used and the number of rows skipped.
5. Prints the computed correlation coefficient r, rounded to 4 decimal places.
6. Classifies and prints the strength/direction of the relationship (e.g. strong positive, moderate positive, weak/none, moderate negative, strong negative) based on standard thresholds on r.

5. Topic: Linear Regression

Dataset

Download the dataset from the link below:
https://drive.google.com/file/d/1NuP8x15x6b_A2NpSCKWTem452JY93V86/view?usp=sharing
The dataset contains 1000 rows and the following 4 columns:

Column	Type	Description
study_time_hours	float	Hours the student spent studying
attendance_percent	float	Class attendance percentage
sleep_hours	float	Average hours of sleep per night
final_exam_score	float	Final exam score out of 100 (target variable)

Question

A researcher wants to predict a student's final_exam_score using three predictors: study_time_hours, attendance_percent, and sleep_hours. Using the dataset provided above, answer the following:

Write a complete C program that:

1. Reads the dataset from the CSV file.
2. Builds the design matrix X (including the intercept column of 1's) and the target vector y.
3. Computes $\beta = (X^T X)^{-1} X^T y$ using matrix transpose, multiplication, and inversion (either the Gauss-Jordan method or the cofactor/adjugate method).
4. Prints the fitted intercept and all three coefficients.
5. Computes and prints the R^2 and RMSE of the model on the full dataset.
6. Predicts and prints the final_exam_score for a new student with study_time_hours = 5, attendance_percent = 90, and sleep_hours = 7.

Submission Guidelines

- Submit your .c source file along with a screenshot of the program's output.
- Code must compile without errors using gcc (C99 or later).
- Add comments explaining each major step of your program.
- Do not use any external statistics libraries — implement the correlation calculation yourself.
- Handle edge cases gracefully: missing values, non-numeric fields, and a zero denominator.